

Andy Tockman

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<https://github.com/tckmn> <https://tck.mn/portfolio>

EDUCATION

Massachusetts Institute of Technology (Cambridge, MA)

2023 – 2025

- Master of Engineering in Electrical Engineering and Computer Science
- GPA: 5.0/5.0

Massachusetts Institute of Technology (Cambridge, MA)

2019 – 2023

- Bachelor of Science in Mathematics
- Bachelor of Science in Computer Science and Engineering
- Minor in Linguistics
- GPA: 5.0/5.0

WORK EXPERIENCE

Lunarch Studios

February 2021 – June 2022

Developer (Full-Time summer 2021, Part-Time otherwise)

remote

- Worked on Islands of Insight, massively multiplayer online puzzle video game
- Created puzzle-authoring interface used for writing thousands of handmade logic puzzles
- Designed internal puzzle database, interfacing with publisher's database for game content
- Implemented spectral-graph-theory-based approach for puzzle generation

TomoCredit

summer 2020

Software Engineering Intern

remote

- Redesigned and implemented new version of landing page (HTML/CSS/JS)
- Fixed bugs and added features in onboarding flow and customer dashboard (JavaScript/React)

Infuse Energy

summer 2017 – 2018

Programming Intern

Houston, TX

- Created web portal with pricing calculator, customer data queries, etc., consolidating and vastly improving speed of existing tools (Python, SQL, HTML/CSS/JS)
- Built mathematical model to predict future energy usage (Python)
- Improved visual appearance of customer email reports (HTML/CSS) including graphs (Python)
- Wrote detailed documentation of all of the above

PUBLICATIONS

Andy Tockman, Pratap Singh, Andres Erbsen, Samuel Gruetter, Adam Chlipala. “Foundational Verification of Running-Time Bounds for Interactive Programs.”

- Presented by coauthors at CPP 2026
- Published in proceedings of 15th ACM SIGPLAN International Conference on Certified Programs and Proofs

MIT Hardness Group, Josh Brunner, Lily Chung, Erik D. Demaine, Della Hendrickson and Andy Tockman. “ASP-Completeness of Hamiltonicity in Grid Graphs, with Applications to Loop Puzzles.”

MIT Hardness Group, Della Hendrickson and Andy Tockman. “Complexity of Planar Graph Orientation Consistency, Promise-Inference, and Uniqueness, with Applications to Minesweeper Variants.”

- Both presented by coauthors at FUN 2024
- Both published in proceedings of 12th International Conference on Fun with Algorithms (2024)

Hugo Akitaya, Josh Brunner, Erik D. Demaine, Della Hendrickson, Victor Luo, and Andy Tockman. “Complexity of Simple Folding Orthogonal Crease Patterns.”

- Presented at TJCDCGGG 2020+1
- Published in Thai Journal of Mathematics special issue: Discrete and Computational Geometry, Graphs, and Games (2023)

TEACHING EXPERIENCE

MIT 6.5210 Advanced Algorithms

Teaching Assistant

fall 2024
Cambridge, MA

- Held office hours to help students with algorithms and abstract problem solving
- Organized peer grading logistics

MIT 6.512 Formal Reasoning About Programs

Undergraduate Teaching Assistant

spring 2023
Cambridge, MA

- Held office hours to help students with formal verification and computer-assisted proofs
- Graded most submissions and gave personalized feedback on proof scripts for each problem set

MIT 18.701 Algebra I

Undergraduate Mentor

fall 2022
Cambridge, MA

- Met with mentees once a week to help with problem sets, proof writing skills, and conceptual understanding

MIT Educational Studies Program

Teacher

periodically 2019 – 2022
Cambridge, MA

- Planned and taught mini-classes such as Complex Analysis, Weird Programming Languages, Code Golf, Cryptic Crosswords, Atomic Chess, Toki Pona, Math in Logic Puzzles, and Surreal Numbers and Games

LEADERSHIP / SERVICE

Tech Squares (MIT square dancing club)

Class Coordinator, Rooming Coordinator

September 2023 – present
Cambridge, MA

- Organized semester-long square dancing classes of ~25 people, responsible for all class-related decisions
- Planned and taught 3 advanced-level and 3 challenge-level classes

ET

Lieutenant Commander, etc.

2019 – 2025
Cambridge, MA

- Organized mealplan, oriented new members, planned events, managed chore distribution system, etc. at MIT independent living group ET

MIT Asymptones (a cappella group)

Musical Director

June 2021 – December 2022
Cambridge, MA

- Ran semesterly auditions and weekly rehearsals, led the group and its musical decisions

MathROOTS (virtual)

Residential Counselor

June 2021
Cambridge, MA

- Assisted with MathROOTS summer program run by MIT PRIMES, for gifted high school students from underrepresented backgrounds (run virtually due to the pandemic)
- Designed and ran daily social events for students for the duration of the program
- Supported students individually with personal/academic counseling

AWARDS

International Linguistics Olympiad

Contestant

2017 – 2019
Dublin, Prague, Yongin

- 2017: silver medalist
- 2018: gold medalist, 2nd place in team contest, best solution for problem #1
- 2019: gold medalist

SKILLS

Languages: English (native), Spanish (intermediate but rusty)

Programming languages: (portfolio of 50+ personal projects: <https://tck.mn/portfolio>)

proficient C, Ruby, Python, Mathematica, HTML/CSS/JavaScript, Haskell, Coq
some experience Rust, Java, C++, SQL
minimal experience x86 assembly, Perl, Julia, R, OCaml

Tools: $\text{\LaTeX}$, git, vim, bash/zsh, nix, basic GIMP/Inkscape (raster/vector graphics)